OOP

1. OOP Basics (Key Concepts for Testing)

Class

Definition: A blueprint or template that defines the attributes (data) and behaviors (methods) of objects.

```
public class Account {
    int accountNumber;
    double balance;
    double getBalance() { ... }
}
```

Object

Definition: An instance of a class. A specific realization of the blueprint.

```
Account myAccount = new Account(); // myAccount is an object
```

Encapsulation

Definition: Bundling data and methods that belong together inside a class, hiding internal details from outside.

Testing Implication: A well-encapsulated class: - Provides a distinct service - Does not depend on other classes - Does not interfere with other classes - All inter-class interaction through **messages**

Inheritance

Definition: A mechanism to move repeated data/methods to a higher-level abstract class, reducing code repetition.

```
Account (abstract parent)
  checkingAccount (child)
  savingsAccount (child)
```

Testing Implication: Abstract classes are NOT executable → NOT directly testable. Solution: **Flattened Classes** (Bob Binder's concept).

Flattened Class: A class that contains all attributes and methods from both itself and its parent class, used specifically for testing.

Abstraction

Definition: Showing only essential features and hiding unnecessary details. Abstract classes define the *what*, not the *how*.

Polymorphism

Definition: The same method name applies to different objects, producing different behavior.

```
// ValidRange is a parent class
// Day, Month, Year, Date all override ValidRange differently
ValidRange
    Day   (overrides and uses parent)
    Month (overrides and uses parent)
    Year  (uses parent as-is)
    Date  (completely overrides parent)
```

Testing Implication: Every polymorphic variant must be covered in class-as-a-unit testing.